

GridWatch OT Detection Lab

A controlled Modbus detection exercise using synthetic traffic and a packet capture; no industrial equipment was connected.

Status: offline PCAP analysis completed

Executive summary

Analyzed a 169-packet synthetic GridWatch capture with Suricata and Zeek. Suricata produced 122 alerts attributed in the lab to attacker source 127.0.0.2. Zeek extracted four TCP connections. The alerts demonstrate protocol-focused detection and support investigation, but do not establish that a real PLC was attacked.

Scope and method

Offline PCAP analysis only. No scanning or traffic was sent to production, public infrastructure, or a live industrial controller.

Results

Alert family	Count / observation
Unauthorized Modbus writes	44 alerts in the simulated capture.
Coil writes / actuator seizure pattern	43 alerts. Signature indicates a suspicious write pattern; it does not show real equipment impact.
Write flood / replay / manipulation	28 alerts. Requires packet and process-context validation.
Read sweep / point enumeration	5 alerts; one diagnostics request and one register-tamper alert were also reported.
Zeek flow context	Four TCP connections extracted from the PCAP.

Evidence screenshots

Figure 1. Splunk source-to-destination alert summary

enterprise

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👤 Administrator

New Search

SPL

Convert to SPL2

source="eve.json" sourcetype="_json"

| stats count values(alert.signature) as alert_signatures by src_ip dest_ip dest_port app_proto

| sort -count

Time range: All time

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✓ 122 events (before 10/4/26 4:11:07:000 PM)

No Event Sampling

Job

|||

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🔦 Smart Mode

Events

Patterns

Statistics (1)

Visualization

Show: 20 Per Page

Format

Preview: On

src_ip	dest_ip	dest_port	app_proto	count	alert_signatures
127.0.0.2	127.0.0.1	5020	modbus	122	GRIDWATCH Modbus coil write (actuator seizure) GRIDWATCH Modbus diagnostics request (recon) GRIDWATCH Modbus holding-register write (setpoint tamper) GRIDWATCH Modbus read sweep (point enumeration) GRIDWATCH Modbus write flood (replay / manipulation of control) GRIDWATCH Modbus write from unauthorized host

Splunk search over the lab eve.json file captured 122 synthetic Suricata events and lists the associated Modbus signatures. This is a screenshot of offline lab analysis, not live OT monitoring.

Figure 2. PCAP generation and Suricata detection summary

```

akachi@akachi-ThinkPad-T440s:~/gridwatch$ make pcap
make detect
uv run python scripts/make_pcap.py capture/gridwatch.pcap
wrote 169 packets to capture/gridwatch.pcap
bash scripts/run_detection.sh capture/gridwatch.pcap
-> Running Suricata over gridwatch.pcap

===== GRIDWATCH DETECTION SUMMARY =====
44 x sid 1000001 GRIDWATCH Modbus write from unauthorized host
43 x sid 1000003 GRIDWATCH Modbus coil write (actuator seizure)
28 x sid 1000020 GRIDWATCH Modbus write flood (replay / manipulation of control)
5 x sid 1000011 GRIDWATCH Modbus read sweep (point enumeration)
1 x sid 1000010 GRIDWATCH Modbus diagnostics request (recon)
1 x sid 1000002 GRIDWATCH Modbus holding-register write (setpoint tamper)

-- Source attribution (alerts must come only from the attacker 127.0.0.2) --
127.0.0.2
=====
Full alert log: /home/akachi/gridwatch/detection/suricata/output/fast.log | Structured events: /home/akachi/gridwatch/detection/suricata/output/eve.json
akachi@akachi-ThinkPad-T440s:~/gridwatch$ zeek -C -r capture/gridwatch.pcap detection/zeek/modbus-detect.zeek
akachi@akachi-ThinkPad-T440s:~/gridwatch$ ls -lh *.log
-rw-rw-r-- 1 akachi akachi 1006 Oct  4 14:24 conn.log
-rw-rw-r-- 1 akachi akachi 278 Oct  4 14:24 packet_filter.log
akachi@akachi-ThinkPad-T440s:~/gridwatch$

```

make pcap wrote 169 synthetic Modbus packets to `capture/gridwatch.pcap`, then the Suricata run reported six GRIDWATCH signatures totaling 122 alerts, all attributed to lab source 127.0.0.2. Terminal capture, Oct 4, 2026, 14:24.

Figure 3. Zeek conn.log from the same capture

```

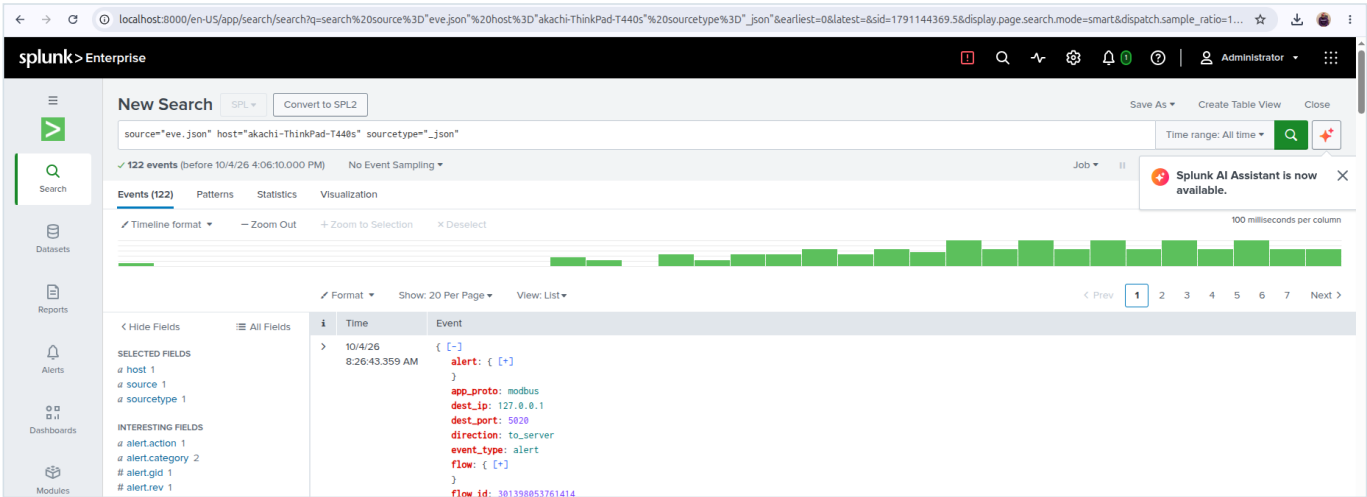
skachi@skachi-ThinkPad-T440s:~/gridwatch$ head -n 25 conn.log
cat packet_filter.log
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path conn
#open 2026-10-04-14-24-11
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p proto service duration orig_bytes resp_bytes conn_state local_orig l
ocal_resp missed_bytes history orig_pkts orig_ip_bytes resp_pkts resp_p tunnel_parents ip_proto resp_bytes conn_state local_orig l
#types time string addr port addr port enum string interval count string count count count count set[stri
ng] count
1600000001.659998 CRBFi81AYWYqjCPLY8 127.0.0.2 50003 127.0.0.1 5020 tcp - 1.719999 492 492 SF T T 0 ShAddFf4
4 2252 43 2212 6
1600000001.439999 CQWQhK1NATAgGr1e4 127.0.0.2 50002 127.0.0.1 5020 tcp - 0.199999 36 36 SF T T 0 ShAddFf6
276 5 236 - 6
1600000000.639999 CEFGrY2tYKGoHiaDD 127.0.0.2 50001 127.0.0.1 5020 tcp - 0.780000 216 309 SF T T 0 ShAddFf2
1 1056 19 1069 6
1600000000.020000 CAWVgLCrdV9snoPw1 127.0.0.1 40001 127.0.0.1 5020 tcp - 0.599999 156 192 SF T T 0 ShAddFf1
6 796 15 792 6
#clear 2026-10-04-14-24-11

```

Zeek parsed the PCAP into four TCP connections to 127.0.0.1:5020 (three from 127.0.0.2 and one from 127.0.0.1), each with SF connection state. These are flow records from the offline capture, not live network sessions. Terminal capture, 14:25.

Evidence screenshots (continued)

Figure 4. Suricata EVE events indexed in Splunk



Splunk returned 122 events from eve.json (sourcetype _json). The expanded event shows a Modbus alert from 127.0.0.1:50003 to 127.0.0.1:5020 with pkt_src wire/pcap, confirming the data came from the offline capture. Splunk capture, 16:06.

Figure 5. Alert count by signature and severity



stats count by alert.signature alert.severity reproduces the 44 / 43 / 28 / 5 / 1 / 1 distribution shown in the Results table. Counts are rule hits, not unique incidents. Splunk capture, 16:07.

GridWatch OT Detection Lab | Analyst notes

Recommended follow-up

1. Review packet timestamps, Modbus functions, source/destination assets, and alert signatures together.
2. Validate whether any write was expected against an authorized engineering source, change ticket, and maintenance window.
3. Escalate through OT operations and safety owners; do not issue controller commands or isolate assets based only on an alert.
4. Tune signatures and thresholds against known-good traffic, then document tested coverage and limitations.

Limitations

All activity was simulated in an offline packet capture. Alert totals are rule hits and can include repeated packets or related signatures; they are not unique incidents. No process state, PLC memory, safety impact, or production availability was measured.

Evidence basis

Terminal evidence recorded the 169-packet capture analysis, 122 Suricata alerts with the lab source attribution, and four Zeek TCP connections. Use the actual capture and logs when reproducing.

Skills demonstrated

Suricata rules and EVE output, Zeek connection analysis, Modbus protocol awareness, PCAP triage, and OT-safe escalation.

Portfolio integrity note: This report describes an educational demonstration. It is not a client engagement, production security assessment, or certification of control effectiveness.